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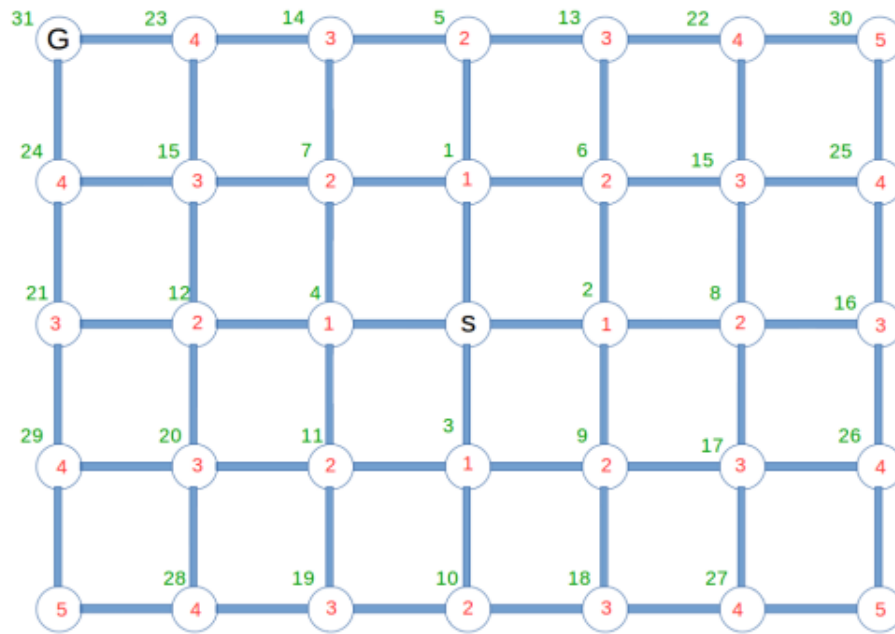
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▼ 1.1 Heuristic Search

1. If the grid is $N \times M$ the state space is also $N \times M$
2. The maximum branching factor is 8 since for example given a starting node with no walls and cells in every direction (N, E, S, W, NE, SE, SW, NW) there are 8 potential cells to branch towards
3. A heuristic that use the Manhattan distance is not admissible since given a wall $\perp$ where the Schrodinger is in the SW, and the fish is in the NE, Schrodinger as a quantum cat can phase through $\perp$ walls in the NE direction therefore there is a true distance of 1 but the Manhattan heuristic would give 2 which is greater than 1 thus inadmissible
4. For the same counterexample as the Manhattan distance a heuristic using Euclidean distance is not admissible as given a wall $\perp$ where the Schrodinger is in the SW, and the fish is in the NE, Schrodinger as a quantum cat can phase through $\perp$ walls in the NE direction therefore there is a true distance of 1. However, the Euclidean distance would give $\sqrt{2}$ which is greater than 1 thus it is also inadmissible
5. Take the difference of the goal vector and the potential new cell, $[goal_x, goal_y] - [x, y]$ then for each element take the absolute value $|goal_x - x|$ and $|goal_y - y|$ then the heuristic is the maximum of the two differences $max(|goal_x - x|, |goal_y - y|)$ this heuristic will not

overestimate the cases described above, since its always a straight line distance on one of the coordinates which is the most optimistic estimate

6. A heuristic for the grid problem described in Paco's notes:



```
G_x, G_y = Goal_point
if |G_y - y| == 0:
    return 0
else:
    return |G_x - x|
```

is admissible since its always less than or equal to the Manhattan distance which is admissible, but is not consistent since if successor point v at $[20,0](x,y)$ is in line in the y coordinate with the goal at $[0,0]$ then v would have value $h(v) = 0$ and so if the original point u is at $[20, 1]$ then it would have value $h(u) = 20$ and $h(u) + h(v) = 20 + 0 \not\leq w(u, v) = 1$. Thus this heuristic is admissible but not consistent

▼ 1.2 Constraint Satisfaction Problem

1. We have 16 variables where $v_{11} \in 1, v_{14} \in 3, v_{23} \in 2, v_{32} \in 1, v_{34} \in 4, v_{42} \in 3$ and the other 10 variables have domains $\{1, 2, 3, 4\}$

2. Constraints:

a. $\forall i, j, k \in [1, 4], v_{ij} \neq v_{ik}$

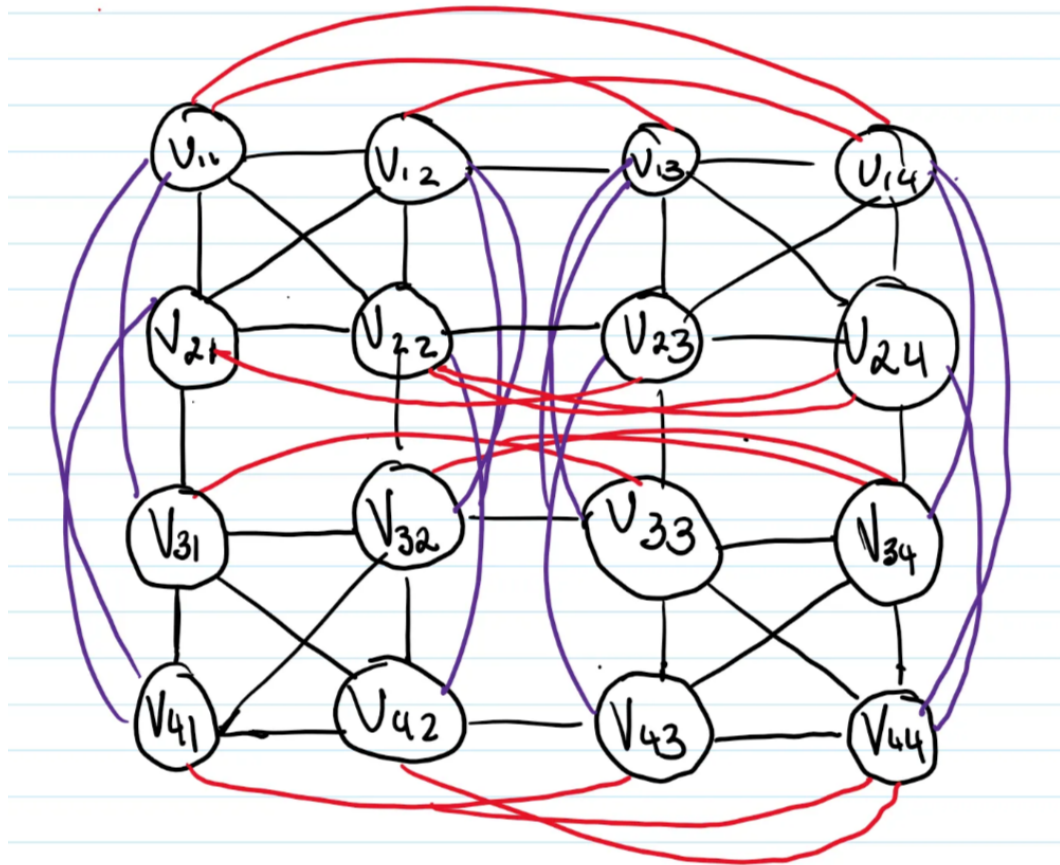
b. $\forall i, j, k \in [1, 4], v_{ji} \neq v_{jk}$

c. $\forall x_{ij}$, such that i, j odd : $x_{ij} \neq x_{i+1, j+1}$ (Top Left)

$\forall x_{ij}$, such that i, j even : $x_{ij} \neq x_{i-1, j-1}$ (Bottom Right)

$\forall x_{ij}$, such that i even, j odd : $x_{ij} \neq x_{i-1, j+1}$ (Bottom Left)

$\forall x_{ij}$, such that i odd, j even : $x_{ij} \neq x_{i+1, j-1}$ (Top Right)



3.

1	2,4	4	3
3,4	4	2	1
2,3	1	3	4
2,4	3	1,4	1,2

4.

5. $v_{13} = 4$ will be chosen since it has the least remaining values and has the most

6. It fails quickly because v_{22} has its only remaining value 4 taken, so it has no more values and the algorithm should fail quick through forward checking